

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **RTL for high-speed Ethernet MAC, PCS, and FEC IP cores**, supporting multi-rate, multi-port architectures up to 1.6Tb.
- Collaborated daily with **design, verification, and back-end engineers** in a co-located team environment to drive project timelines and resolve technical challenges.
- Learned and applied **digital design best practices** including synthesis, timing closure, and configurable IP delivery for complex electronic systems.
- Developed **Python scripts** to automate data pipelines and scheduled reporting, enhancing operational efficiency by **15%**.
- Managed production support for enterprise applications, responding to P1/P2 incidents and ensuring adherence to strict **SLAs**.
- Developed and maintained **REST APIs** using Java Spring Boot for inter-service communication, supporting seamless integration across enterprise systems.

SKILLS

Hardware Description Languages – Verilog, SystemVerilog

Digital Design – RTL Design, State Machines, Clocking, Timing Closure, Synthesis, FPGA Implementation

Networking – Ethernet Standards (IEEE 802.3dj), High-Speed Networking

Tools & Methodologies – Xcelium, VCS, ModelSim, Jenkins, Git, Jira, Confluence, Docker

Programming Languages – Python, Java, SQL

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch)

Data Analysis & BI – Power BI, DAX, Power Query, Tableau, R, pandas, NumPy, scikit-learn

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

High-Speed Network Interface Controller IP

- Designed and verified **RTL modules for a multi-gigabit Ethernet MAC**, adhering to IEEE standards for network performance.
- Implemented **configurable parameters** within the IP core to support flexible port configurations and data rates.
- Collaborated with verification engineers to ensure functional correctness and achieve **timing closure** targets under simulated load conditions.

Automated Data Processing Pipeline

- Developed a **Python-based ETL pipeline** to ingest and transform large datasets from various sources into a centralized data warehouse.
- Integrated **SQL queries** for data validation and cleansing, improving data accuracy for downstream analytics by **98%**.
- Scheduled and monitored pipeline execution using **AWS Lambda and CloudWatch**, ensuring reliable daily data updates for business intelligence reports.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)